

SYSTEMS AND METHODS FOR LONGITUDINAL PELVIC PHYSIOLOGY MONITORING USING A MULTI-MODAL INTRALUMINAL PHYSIOLOGIC MONITORING DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims priority to and is a continuation-in-part of prior U.S. provisional applications directed to distributed physiologic sensing platforms comprising multiple sensing nodes operating individually or cooperatively within a computational analysis system. The longitudinal pelvic physiology monitoring device described herein may operate independently or as a functional node within a multi-device biometric ecosystem configured for longitudinal physiologic analysis, adaptive calibration, compatibility modeling, personalized physiologic profiling, relational physiology analysis, and health intelligence modeling. All related disclosures are incorporated herein by reference in their entirety.

FIELD OF THE INVENTION

The present disclosure relates generally to biometric monitoring systems and physiologic sensing technologies. More particularly, the disclosure relates to systems and methods for longitudinal monitoring of pelvic and intraluminal physiology using a body-positioned intraluminal physiologic monitoring device configured for temporary insertion, episodic measurement, retained placement, or repeated use within a body lumen. In various embodiments, the device is configured to obtain physiologic signals associated with tissue interaction, hydration state proxies, dielectric properties, temperature context, pressure distribution, compliance behavior, motion state, and related biologic conditions while maintaining user comfort and consistent positional association during measurement. The disclosure encompasses implementations suitable for vaginal, anal, or other intraluminal physiologic environments without limitation to specific anatomical implementations.

BACKGROUND OF THE INVENTION

Existing pelvic monitoring technologies are predominantly episodic, clinic-based systems relying on wired probes, imaging equipment, or rigid diagnostic tools unsuitable for longitudinal physiologic observation outside clinical environments. Such systems fail to capture dynamic physiologic variability occurring across daily life, circadian cycles, hormonal variation, recovery states, and environmental context. There exists a need for a non-intimidating intraluminal physiologic monitoring device capable of acquiring repeatable longitudinal data while preserving privacy, comfort, and accessibility outside clinical settings. The present disclosure addresses these limitations through a multi-modal intraluminal sensing platform capable of continuous, periodic, or session-based physiologic measurement and conversion of physiologic signals into abstracted indices suitable for computational modeling, personal insight systems, or distributed physiologic analysis platforms.